



**Open source Nexus modelling tools for
Planning sustainable Energy Transition in Africa**

**D2.3 Design, development and deployment of ONEPlanET database.
Explanatory note**

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Acronyms and abbreviations table

Acronym	Meaning
AFREC	African Energy Commission
CC	Capital Compensation
CS	Case Study
D	Deliverable
dmnl	Dimensionless
DN	Digital Numbers
EN-D2.3	Explanatory Note of D2.3
GDP	Gross Domestic Product
GFCF	Gross Fixed Capital Formation
GHSL	Global Human Settlements
GIS	Geographical Information System
HRSL	High-Resolution Settlement Layer
IEA	International Energy Agency
IUCMA	Inkomati Usuthu Catchment Management Agency
IUWMA	Inkomati Usuthu Water Management Area
LCCS	Land Cover Classification System
MP	Model Parameters
NTL	Night-Time Light
RES	Renewable Energy Sources
SD	System Dynamics
SSP	Shared Socioeconomic Pathways
T	Task
TFEC	Total Final Energy Consumption
TLSP	Taxes Less Subsidies on Products
WP	Work Package

1. Executive Summary

This Explanatory Note for the Deliverable 2.3 (EN-D2.3) outlines the design, development, and deployment of the database created for the ONEPlanET Project. Deliverable 2.3, which is the Excel File attached to this note and represents a key bridge between the data-gathering phase carried out in this WP2 and the subsequent modelling efforts in WP3 and beyond.

The core purpose of this document is to provide a comprehensive account of the database-related activities performed under Task 2.5. It details the full workflow, from initial conceptual design through to data collection, processing, validation, and integration, resulting in a structured, Excel-based database designed to feed into System Dynamics (SD) models developed using Vensim DSS. This modular database covers four interconnected sectors: Water, Energy, Food & Land, and Socio-Economy. Each module is mapped to its own tab in the Excel file.

Data sources include both national and geolocated data, ensuring breadth and contextual specificity. Where granular data were unavailable, downscaling methodologies have been implemented, most notably with Gross Domestic Product (GDP) as downscaling driver, to estimate values at river basin level. The Inkomati-Usuthu Water Management Area (IUWMA) served as the test case for applying and validating these methodologies.

Metadata has been embedded directly into the database files to facilitate transparency and reproducibility. Despite challenges related to data availability and geographical resolution, they have been successfully mitigated through methodological flexibility, the use of GIS, and iterative collaboration with modelling partners.

2. Introduction

2.1. Purpose of the document

This Explanatory Note of Deliverable 2.3 (EN-D2.3) summarises all the activities carried out under Task 2.5 (T2.5) and is framed in Work Package 2 (WP2), being the last report of this WP. This task serves as a liaison between WP2 and WP3, as here the necessary data to populate the modules are collected in an iterative exercise that engages the interconnection among the modelling teams and the partners searching for data.

This file aims to report on the whole data-related process, encompassing the definition and design of the database, the mapping and searching of datasets and sources, the further data collection and, eventually their processing and adaptation to the most suitable format. In parallel, this document also addresses some of the activities requested by the Grant Agreement (GA), such as the inclusion of metadata, and the identification of barriers and solutions to data gathering.

D2.3 consists of the Excel file that gathers all the data and metadata necessary to make the model work, and this EN-D2.3 is created to facilitate its understanding and to keep track of the whole creation process. It is made up of several different sections, starting with the database conceptualisation, where key aspects such as the architecture, the data to be included or the format in which it will be released (Excel file, CSV, SQL...) are described. In addition, this section brings together the data requested by each modelling team to feed their modules.

The following chapter focuses on listing the data sources consulted, both those provided by Task 2.1 and other new ones essential to fulfil the modellers' requirements. One of the most critical points of this task, data downscaling, is tackled here by applying different methods (depending on the variable) and their subsequent validation. Metadata is then discussed to determine what kind of information and how it should be integrated into the database to ensure that it is properly contextualised and understood. Finally, the last section deals with the roadblocks encountered during the data search and how they have been overcome.

2.2. Relation to other activities in the project

ECREEE is the WP leader and will supervise the successful implementation of all the tasks of WP2. In addition, CARTIF is responsible for the design and development of the project's database (Task 2.5), taking care of populating in with sectoral data, metadata and algorithms of downscaling and/or upscaling. The rest of partners will collaborate on searching for datasets, contacting model developers and populating the database.

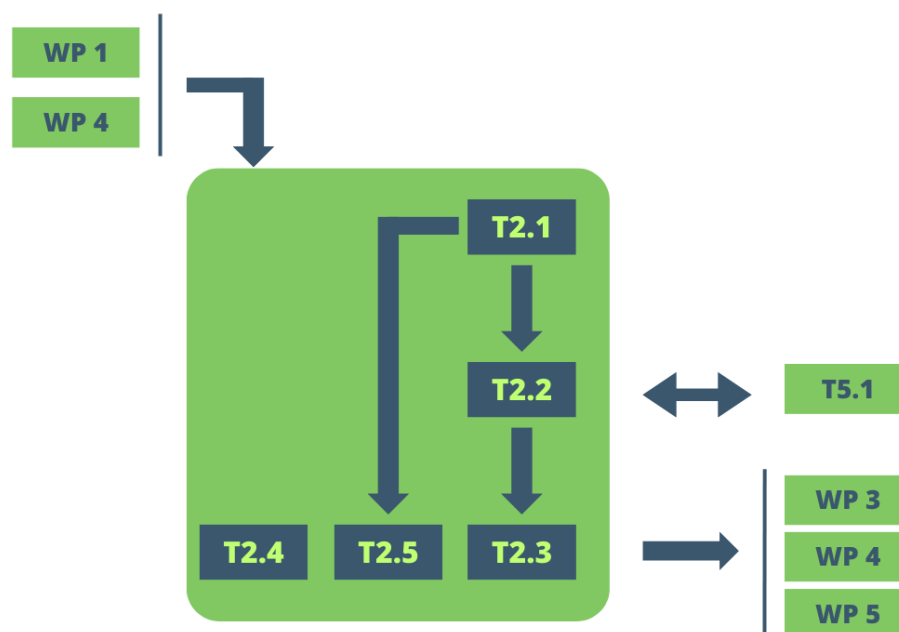


Figure 1. WP2 tasks interaction. Source: Own elaboration.

WP2 receives inputs from WP1 and WP4 and produces outputs that are used for the development of the tasks within WP3, WP4 and WP5. As shown in **Figure 1**, Task 2.1 feeds Task 2.5 with global, national and sub-national data sources.

2.3. Contribution of partners

In total, eight partners have been involved in T2.5. The table below (**Table 1**) describes their contribution to this deliverable:

Table 1. Contribution of partners

Participant name	Contribution to D2.3
CARTIF	Deliverable leader. Coordination and participation in different sections.
ECREEE	Search for data for the Bani CS.
CYI	Search for energy databases.
IHE Delft	Search and validation of water data.
UKZN	Search and validation of land and food data.
RINA-C	Participation in sections 3 and 4.
UVa	Search and validation of energy and economy data.
SEI	Search and validation of water data.

3. Database design and structure

In the design of the database, two main input sources have been considered. On one hand, the information collected in WP2, particularly through the sectoral conceptual models, both the initial common models and the versions adapted to each Case Study (CS) through the co-design workshops (in Task 2.3, as reported in D2.2). On the other hand, the data requirements identified from the modelling tasks in WP3 (Task 3.1 and 3.2, reported in D3.1 and D3.2) have also been taken into account. This process of data identification and modelling has been carried out iteratively: the data required by the models was sought, and the models were adjusted when the necessary data was not available.

In general, the data structure is common across the three CSs. As mentioned, when possible, specific elements collected during the consultation processes have been incorporated. For example, each case study includes data on its predominant crops, as well as on its renewable energy sources (e.g., geothermal energy in the case of Songwe).

3.1. Database concept

The database serves as the primary data source for the System Dynamics (SD) model developed under WP3. Its main purpose is to organize, centralise, and structure the quantitative inputs required by the models in a transparent, reproducible, and flexible way.

Given the diversity of data types and sources involved, it was prioritised the interoperability between the database and the modelling environment. The software used to build up the models, Vensim DSS, supports multiple input formats and, after discussion and agreement with the modelling teams, it was decided to establish a direct linkage between the models and Microsoft Excel spreadsheets. This format enhances:

- Transparency: Allows quick visibility of units, metadata, sources and general structure of the database (tabs, submodules, etc.)
- Flexibility: Easily modifiable, updatable and replaceable without the need of recoding the models or using additional software.
- Accessibility: Easy to use and understand for all partners and external users.
- Traceability: Version control and structured nomenclature conventions allow clear data tracking and updates over time.
- Versatility: Excel files are directly convertible to Python, which is the programming language required by the simulation tool. Additionally, data is vectorised, making programming more efficient.

3.2. Database architecture

The model is structured into four distinct yet interconnected modules: Water, Energy, Food-Land, and Socio-Economy; that operate simultaneously as part of an integrated system. As stated in the “Common Rules for Modelling” document produced by UVa, each module is linked to a dedicated Excel tab within the common file named according to the sector it represents and the associated CS (e.g. Bani Water_MP (Model Parameters)), which acts as its primary data repository. Thus, the Excel file is organised with 15 “data sheets”, each corresponding to a specific module and CS: Energy, Water, Land, Economy and Population for Inkomati-Usuthu (IUWMA),

Songwe and Bani. Additionally, an overview tab per CS with general information of interest is included, e.g. population, inhabited area, access to electricity or involved countries.

The structural design of the database ensures a modular but cohesive approach, where each sectoral module operates with its own dataset while maintaining dynamic interactions with other modules. These interactions allow for the exchange of key variables, ensuring consistency in the simulation process and enabling a comprehensive analysis of interdependencies within the nexus framework.

3.3. Sectoral modules: Data requirements

The historical datasets compiled for this study aims to span from 2010 to 2020, depending on data availability. Each variable is recorded annually, ensuring a consistent temporal resolution, while the spatial resolution is maintained at the river basin level to align with the study's geographical scope. It is noteworthy, however, that some parameters are assumed to be equal to national ones, due to the impossibility of accessing to more accurate data, e.g. labour share, water consumption rates per energy technology, etc.

Data is stored in structured Excel sheets following a standardised format, including in the first columns the essential metadata such as the variable name, measurement units, data source and a brief description of the meaning of each parameter and its calculation methodology. The following columns contain the quantitative data that feed the model, organised on an annual basis. At the end of each Excel file there is table listing all the data sources and a hyperlink to the original online datasets. Some variables require further disaggregation, such as agricultural data classified by crop type or energy consumption categorised by source. To maintain this level of detail, the database includes multiple rows of each sub-type, ensuring consistency across sectoral analyses.

The integration of these datasets into the SD model is facilitated through Vensim DSS's '*Get Direct Data*' and '*Get Direct Constants*' functions. These functions retrieve data directly from the Excel files by requiring a specified file path, sheet name, and defined row and column references. To simplify this process, named ranges are assigned within Excel, making data calls more efficient and reducing the risk of errors.

To meet the needs of the modelling teams, the data should be standardised in terms of temporal and geographical coverage, with historical records complemented by projections where applicable. Variables should be expressed in consistent units, and across-sectoral data must be harmonised to preserve coherence across different modules. The structured format of the dataset ensures seamless integration with Vensim DSS, supporting accurate simulations and reliable analytical outcomes.

An overview of the variables identified preliminarily for each sector is presented below (**Table 2**):

Table 2. Overview of the preliminary data

Module	Number of preliminary variables
Energy	16
Water	21
Food & land	11
Economy	11
Population	5
Total	64

As indicated, this represents a preliminary list of variables. During the development of the models, some of these variables were discarded, while others do not appear explicitly in the model but have been used as intermediate variables for the calculation of other model parameters. Therefore, this document/database aims to harmonize the data collection process and serve as a repository for storing the historical data used to feed the model.

3.3.1. Energy Module

The requirements of the energy module focus on critical points to be considered, as energy is one of the main pillars of the project. They aim to outline how people in the basins consume and produce energy according to specific criteria. In this sense, the majority of the requirements are related to the energy demand (either from households or specific sectors), energy generation metrics, energy potential, and other minor requirements. Disaggregation occurs based on location or by energy type and the required data need to be measured annually for several historical years. The data relevance is high in the majority of the cases except in a few contexts. **Table 3** illustrates the requirements identified by the energy modelling team:

Table 3. Data requirements from the Energy module

Module	Req. Data	Disaggregation	Time	Units	Data Relevance
Energy	Energy demand of households	By final energy type By region	2013-2020	J/yr	High
	Households	By region	2013-2020	Households	High
	Energy demand of productive sectors	By final energy type By region By sector	2013-2020	J/yr	High
	Installed capacity	By technology By region By status	Initial (2020) Under construction Planned	MW	Medium
	Actual age of Installed capacity	By technology By region	2020	yr	High
	Water consumption factor for energy production	By technology By region	2020	m ³ /MW	Medium

	Fuel extraction water factor	By fuel By region	2020	m ³ /MW	Medium
	CO ₂ emissions factor	By technology By region	2020	kg CO ₂ /GJ	Medium
	Grid extension	By region	2020	km	Medium
	Potential for RES	By RES By region	2020	MW	High
	Net energy imports	By final energy type By region	2013-2020	J/yr	High
	Capacity factors	By technology By region	Monthly if possible	dmnl	Medium
	Land occupation factor	By technology By region By status	Initial (2020) Under construction Planned	km ² /MW	Medium
	Space for dams	By region	2020	km ² m ³	Medium
	Built-up area	By region	2010-2020	km ²	Low
	Forest area	By region By type of forest	2010-2020	km ²	Low

The technologies considered for the data collection are solar PVs, wind, hydropower, biomass, coal, gas and oil, while the productive sectors are industry, mining, agriculture, transport and services.

Since the modelling work is still ongoing, it is possible that some variables included in the final models may not appear in this initial list, either because they have not been modelled directly or because they have served as auxiliary variables used in the calculation of other parameters. For example, the energy model includes the variable “energy intensity”, which is calculated based on the Total Final Energy Consumption (TFEC) divided by the corresponding sectoral GDP.

3.3.2. Water Module

The data requested by this module can be classified into three major groups: climate-related data, available waterbodies, and water demand from various sources. Thus, the water module needs monthly data such as rainfall, evaporation or runoff to calculate water availability as well as the historical trends of water volume in different reservoirs, e.g. springs, lakes, groundwater storages... Finally, to model the entire water cycle, withdrawals in form of demands from sectors like farming, livestock, industry or mining, are also requested with high priority. This latter request should be disaggregated per crop, livestock class and productive sector. **Table 4** brings together water module’s data requirements:

Table 4. Data requirements from the Water module

Module	Req. Data	Disaggregation	Time	Units	Data Relevance
Water	Rainfall	By region	Monthly if possible 2010-2020	mm	High
	Interception losses	By region	Monthly if possible 2010-2020	mm	High
	Infiltration	By region	Monthly if possible 2010-2020	mm	High
	Evaporation & Evapotranspiration	By region	Monthly if possible 2010-2020	mm	High
	Runoff	By region	Monthly if possible 2010-2020	mm/s	High
	Soil moisture	By region	Monthly if possible 2010-2020	m ³	High
	Groundwater storage	By region	2010-2020	m ³	High
	Water in wells	By region	2010-2020	m ³	High
	Water in boreholes	By region	2010-2020	m ³	High
	Water in springs	By region	2010-2020	m ³	High
	Water in lakes	By region	2010-2020	m ³	High
	Water in rivers	By region	2010-2020	m ³	High
	Water in reservoir	By region	2010-2020	m ³	High
	Environmental water requirements	By region	2010-2020	m ³ /yr	Medium
	Forestry water requirement	By region	2010-2020	m ³ /yr	Medium
	Domestic water demand	By region	2010-2020	m ³ /yr	High
	Water demand by sector	By productive sector By region	2010-2020	m ³ /yr	High
	Livestock water demand	By livestock class By region	2020	L/head	Medium
	Livestock heads	By livestock class By region	2010-2020	head/yr	Medium
	Crop water requirements	By crop By region	2020	L/ha	Medium
	Harvested area	By crop By region	2010-2020	ha/yr	Medium

3.3.3. Food & Land Module

The most relevant data for this system are the land use types in each CS, since each class has specific usages and characteristics (forests, urban land, water bodies, croplands, etc.) In particular, it is a priority to collect accurate data on cattle and on how crops are harvested (yields and extension) to estimate food production, water consumption, and emissions. Additionally, other information useful for modelling food demand (population) and carbon flows (biomass) is needed. All data are requested on an annual basis, as reflected in **Table 5**.

Table 5. Data requirements from the Food & Land module

Module	Req. Data	Disaggregation	Time	Units	Data Relevance
Food & Land	Land Class Cover	By land use type By region	2010-2020	ha	High
	Harvested area	By crop type By region	2010-2020	ha	High
	Livestock heads	By livestock class By region	2010-2020	head/yr	High
	Crop yields	By crop type By region	2010-2020	kg/ha	High
	Land use change emissions	By region	2010-2020	MtCO ₂ eq/yr	Medium
	Livestock emissions	By livestock class By region	2010-2020	kgCH ₄ / head·yr	Medium
	Cropland emissions	By region	2010-2020	tCH ₄ /yr	Medium
	Forest Loss	By region	2010-2020	ha	Medium
	Aboveground biomass	By region	2010-2020	t/ha	Medium
	Belowground biomass	By region	2010-2020	t/ha	Medium
	Population	By region	2010-2020	People	High

3.3.4. Socio-Economy Module

Since each component of this module differs considerably from the other, their data requirements have been split into two separate sub-sections.

Economy Module

The economy module aims to clearly portray the economic and financial context by collecting data related to the floridity and power of the investments done within the basins. This module includes a massive variety of requirements, in particular, GDP, household expenditure, Government spending, net exports, labour share, investments, employment, but also resources needed and available by sector. The approach to these data begins with the downscaling of regional or national GDP to the river basin level (explained in **Section 4.2.1**) and then using it as vector to downscale the other required data (explained in **Section 4.2.3**). In some cases, the disaggregation is also done by productive sector. The data for the analysis is collected on an

annual basis for different years within the historical period. **Table 6** shows the requirements identified by the economy modelling team:

Table 6. Data requirements from the Economy sub-module

Module	Req. Data	Disaggregation	Time	Units	Data Relevance
Economy	Annual GDP in constant prices	By region By productive sector	2010-2020	Const. US Dollar	High
	Households final consumption expenditure	By region	2010-2020	Const. US Dollar	High
	Government spending	By region	2010-2020	Const. US Dollar	High
	Net exports	By region	2010-2020	Const. US Dollar	High
	Gross operating surplus	By region	2010-2020	Const. US Dollar	High
	Labour share	By region	2010-2020	% of GDP	High
	Depreciation rate	By region	2010-2020	% of GDP Const. US Dollar	High
	Gross fixed capital formation	By region	2010-2020	Const. US Dollar	High
	Taxes less subsidies on production	By region	2010-2020	Const. US Dollar	High
	Capital stock	By region	2010-2020	Const. US Dollar	High
	Capital Compensation	By region	2010-2020	Const. US Dollar	High

Demography/Population Module

This module's requirements include one of the most significant data, which is the population located in the river basins. Population should be geolocated with the highest possible resolution and accuracy because it will serve as a basis for downscaling several variables in other modules. Besides inhabitants, fertility, mortality and migration rates are also demanded from the population module, but the approach to these data follows a national perspective, assuming homogeneity across the country because of the lack of accessibility to sub-national sources. Finally, household composition is also requested to calculate consumption rates.

The highest priority data are population, and mortality and fertility rates, which have to be broken down by gender and age groups. **Table 7** shows the requirements identified by the corresponding modelling team:

Table 7. Data requirements from the Population sub-module

Module	Req. Data	Disaggregation	Time	Units	Data Relevance
Population	Initial Population	Yearly and gender	2010-2022	people	High
	Fertility Rate	Yearly and gender	2010-2022	per 1000 people	High
	Mortality Rate	Yearly and gender	2010-2022	per 1000 people	High
	Migration Rate	-	2010-2022	per 1000 people	Medium
	Household Composition	-	2015-2022	people	Medium-high

3.3.5. Projections

Beyond historical data, the model also needs exogenous forecasts of some variables to create potential future scenarios. These estimations should be downscaled to the CS level, but they originate from the Shared Socioeconomic Pathways (SSP) projections, a set of pre-defined scenarios that describe how the world might evolve under different socioeconomic assumptions. Some of them are available geolocated, but others are only national, so a further disaggregation has to be performed (as explained in **Section 4.2**). **Table 8** contains these projections:

Table 8. Required data projections

Module	Req. Data	Disaggregation	Time	Units	Data Relevance
Cross-modal	Population	By region	2020-2100	people	High
Water	Rainfall	By region	2020-2100	mm	High
Water	Temperature	By region	2020-2100	°C	High
Cross-modal	Emissions	By GHG By region	2020-2100	MtCO ₂ eq/yr	Medium
Economy	GDP	By region	2020-2050	US Dollar	Medium

4. Data collection and sources

4.1. Data sources

The data collection process has been conducted in accordance with the requirements established by the modelling teams and the specific needs of the model. Each sectoral module necessitates a defined set of data with a specific temporal and spatial resolution, and in cases where further disaggregation is required, datasets must include relevant subcategories to ensure a detailed and accurate representation of the system.

Two different approaches have been followed depending on the data availability.



When **national / large-scale data** is available (sources identified in Section 4.1.1) → downscaling process based on different criteria, as it will be described in Section 4.2.



When **geolocated data** is available (sources identified in Section 4.1.2) → computing at River basin scale

4.1.1. Large-scale sources

When river basin specific or geospatial datasets are unavailable for certain variables, the next best approach is to use large-scale national or regional datasets, which can be later downscaled to the river basin level. The reliability of the downscaled results is directly linked to the accuracy and resolution of the original dataset. For example, provincial datasets provide more precise estimations than national data, while national datasets offer better insights than continental or global sources. In cases where no national or regional data are accessible, downscaling from continental or global databases remains an option, although it introduces higher levels of uncertainty. The methodologies applied for data downscaling, along with their limitations, are detailed in Section 4.2.

Some of the data sources reported in **Table 9** come from the work done in T2.1 where local partners provided with datasets of interest, while the rest have found after reviewing international organisation and national or sub-national authorities.

Table 9. Large-scale data sources used in ONEPlanET

Database	Time period	Module	Variable	URL
AFREC - Energy Balances	2000-2022	Energy	Final Energy Consumption by sector and type	https://au-afrec.org/data-statistics-energy-balances
IEA – Energy Systems	2000-2022	Energy	Final Energy Consumption by sector and type	https://www.iea.org/countries/
South African Coal Sector Report	2020	Energy	Coal Reserves (IUWMA)	https://www.dmre.gov.za/Portals/0/Energy_Website/files/media/explained/South-African-Coal-Sector-Report.pdf
CSIR Energy Centre	2022	Energy	Capacity Factors	https://www.csir.co.za/sites/default/files/Documents/Statistics%20of%20power%20in%20SA%2022-CSIR-%5BFINAL%5D.pdf
AQUASTAT	2010-2022	Water	Water consumption per capita	https://data.apps.fao.org/aquastat/?lang=en
FAO	-	Water	Crop water requirements	https://www.fao.org/land-water/databases-and-software/crop-information/en/
FAO	-	Water	Livestock water requirements	https://openknowledge.fao.org/server/api/core/bitstreams/2a30f51b-ce97-4e18-a8cf-e80ea0e8892c/content
World Bank Group	2010-2022	SocioEconomy	GDP	https://data.worldbank.org/indicator/NY.GDP.MKTP.KD?name_desc=false
IMF DATAMAPPER	2010-2022	SocioEconomy		https://www.imf.org/external/datamapper/NGDP_RPCH@WEO/OEMDC/ADVEC/WEOWORLD
UN National Accounts	2010-2020	SocioEconomy	Government & Household final consumption	https://unstats.un.org/unsd/snaama/
Economic Data - FRED	2010-2022	SocioEconomy	Capital Stock	https://fred.stlouisfed.org/categories/33405
ILO – Modelled Estimates	2010-2022	SocioEconomy	Labour Share Compensation of employees	https://ilostat ilo.org/topics/sdg/
Economic Annual Report – Malawi	2010-2020	SocioEconomy	GDP TLSP GFCF Capital Stock Net Exports Gross operational surplus Depreciation	https://nsomalawi.mw/publications/economy/
National Accounts of Tanzania	2011-2020	SocioEconomy		https://www.nbs.go.tz/nbs/takwimu/na/National_Accounts_of_Mainland_Tanzania_2020.pdf
Mpumalanga gross domestic product report	2013-2022	SocioEconomy		https://www.statssa.gov.za/publications/D04411/D044112022.pdf
Comptes Nationaux Annuels – Anstat (Côte d'Ivoire)	2015-2022	SocioEconomy		https://www.anstat.ci/assets/publications/files/COMPTE_NATIONAUX_ANNUELS_DEFINITIFS_2022.pdf

Comptes Nationaux – INSTAT (Mali)	2015-2022	SocioEconomy		https://www.instat-mali.org/fr/publications
South Africa Reserve Bank Full Quarterly Bulletin	2015-2022	SocioEconomy	Government & Household final consumption GFCF Capital Stock Net Exports Gross operational surplus Depreciation	https://www.resbank.co.za/content/dam/sarb/publications/quarterly-bulletins/quarterly-bulletin-publications/2024/december-2024/01FullQuarterlyBulletin.pdf
UN – Population Database	2010-2020	SocioEconomy	Age-specific fertility rates by single age Female Mortality Rate by single age Male Mortality Rate by single age Sex ratio at Birth	https://population.un.org/wpp/downloads

4.1.2. Geolocated sources

When site-specific data are not available within the boundaries of the case study, geolocated datasets provide an alternative by offering high-resolution spatial information in grid-cell formats, typically ranging from 100 meters to several kilometres in size. These datasets require additional data processing to adjust their scale to the river basin level, a task generally performed using Geographical Information System (GIS) software such as *ArcGIS* and *QGIS*. The specific methodologies used for spatial data processing and adaptation are further elaborated in Section 4.2.2.

In cases where geolocated datasets are unavailable, the most reliable and precise alternative comes from river basin specific databases, which are typically managed by local organizations and agencies responsible for on-site data collection and monitoring. These institutions, such as the *Inkomati-Usuthu Catchment Management Agency (IUCMA)*, maintain high-resolution datasets that provide valuable insights into local, hydrological, environmental, and socio-economic conditions. However, access to such databases is often restricted due to their private nature, requiring direct negotiation with the respective agencies to obtain the data. In the specific case of the IUWMA river basin, these datasets could not be incorporated into the study due to a lack access to the necessary information.

GIS repositories store geospatial data either raster or vector-based formats. Raster datasets consist of grid cells (pixels), where each cell is associated with specific attribute values corresponding to the measured variable. Vector-based data, on the other hand, represents spatial features through points, lines or polygons, linking geographic coordinated with attribute information. The spatial resolution of a GIS dataset, typically expressed in meters or degrees, determines the level of detail in representation; smaller grid cells provide higher accuracy but require greater computational resources for processing and analysis.

To visualize, analyse, and process geolocated data, this study utilised *QGIS Desktop 3.34.7*, a widely recognised open-source GIS platform. The integration of GIS data into the modelling framework enables a more refined spatial analysis, ensuring that sectoral modules operate with geospatially coherent datasets. This approach enhances the accuracy of model outputs by incorporating high-resolution data tailored to the specific characteristics of each river basin.

GIS databases utilised in this project's data collection process are listed in **Table 10**:

Table 10. Geolocated databases used in ONEPlanET

Database	Time period	Module	Variable	URL
Global Solar Atlas	2025	Energy	Solar PV Potential	https://globalsolaratlas.info/download/south-africa (The World Bank, 2025)
Global Wind Atlas	2025	Energy	Wind Potential	https://globalwindatlas.info/en/ (The World Bank, 2024)
Global Dam Watch	2025	Energy	Hydropower Potential Dammed Water Availability	https://www.globaldamwatch.org/ (Lehner et al., 2024)
WRI - Global Power Plant Database	2024	Energy	Installed Capacity Under construction Capacity	https://www.wri.org/research/global-database-power-plants (Global Energy Observatory (GOE) et al., 2021)
GEM - Global Integrated Power Tracker	2025	Energy	Projected Capacity Fossil Fuel Reserves	https://globalenergymonitor.org/projects/global-integrated-power-tracker/
CWatM v1.04	2020	Water	Water Demands Surface & Groundwater	https://gmd.copernicus.org/articles/13/3267/2020/ (Burek et al., 2020)
HYDRORIVERS	2024	Water	Annual surface water flow	https://www.hydrosheds.org/products/hydrorivers (Lehner & Grill, 2013)
HYDROLAKES	2024	Water	Available water in lakes	https://www.hydrosheds.org/products/hydrolakes (Messenger et al., 2016)
WHYMAP	2023	Water	Groundwater Availability	https://www.whymap.org/whymap/EN/Home/whymap_node.html
Copernicus - LCCS	1992-2022	Food & Land	Land Cover Classes	https://cds.climate.copernicus.eu/datasets/satellite-land-cover?tab=download (Copernicus Climate Change Service, 2019)
MAPSPAM	2000, 2005, 2010, 2017, 2020	Food & Land	Harvested Area Crop Yields Crop Productions	https://mapspam.info/data/ (IFPRI, 2016, 2019, 2020, 2024)
Global Forest Watch	2001-2023	Food & Land	Tree Cover Lost Forest-related Emissions	https://www.globalforestwatch.org/ (Hansen et al., 2013) (Harris et al., 2021)
EDGAR 2024	2010-2023	Food & Land	CH ₄ Emissions	https://edgar.jrc.ec.europa.eu/gallery?release=v2024ghg&substance=CH4&sector=TOTALS
GLW - Gridded Livestock Density	2010, 2015, 2020	Food & Land	Livestock Heads	https://dataverse.harvard.edu/dataverse/glw (Gilbert et al., 2022)

CEDA ARCHIVE - ESACCI	2010, 2015- 2021	Food & Land	Aboveground Biomass	https://data.ceda.ac.uk/neodc/esacci/biomass/data/agb/maps/v5.01/geotiff/aggregated
WorldPop	2000-2020	SocioEconomy	Population	https://hub.worldpop.org/geodata/listing?id=75
SSPs Population Projections	2020-2100	SocioEconomy	Population projections	https://figshare.com/articles/dataset/Projecting_1_km-grid_population_distributions_from_2020_to_2100_globally_under_shared_socioeconomic_pathways/19608594/2 (Wang et al., 2022)
PopClusters	2016	SocioEconomy	Population Electricity Access Night-time light (NLT) Area	https://data.mendeley.com/datasets/z9zfzk8cr/6 (Khavari, 2020)

4.2. Downscaling algorithms

As described in the introduction to this section, many of the databases collected for this project are at the national/regional scale, provided either by national statistical departments or international organisations. Although these data are accurate and totally valid, they do not meet the scale requirements to be achieved by ONEPlanET models, which should at least be at basin level. Therefore, it is imperative to downscale this national information to a more detailed scale and, to this end, there are several key variables that will be used as “downscaling drivers”, i.e. parameters to be downscaled with the highest possible resolution and precision which will then be utilised to rescale the rest of national variables.

Essentially, three downscaling drivers can be used to downscale all other variables: GDP (economy module), population (demography module) and area (food & land module). These three variables are easily accessible even at sub-national level, especially the population and surface of some cities and areas. In fact, these two drivers are available in a geolocalised way in WorldPop (WorldPop, 2020), PopClusters (Khavari, 2020) and LCCS (Copernicus Climate Change Service, 2019) datasets, so there is no need for downscaling them, but for processing the data to aggregate them at the basin and sub-basin levels.

Regarding GDP, the most accurate available data are at the provincial in the IUWMA CS and at country level for the Songwe and Bani CSs, but these levels of detail are not enough. Although they can offer a holistic view of the average economic condition, they are not precise in terms of regional specificities, rural-urban population ratios, etc., so further refinement is needed.

Sub-sections below present the process followed to downscale GDP, and aggregate gridded population and area to the river basin level. The described procedure reflects **the IUWMA CS example, but it is perfectly replicable for the other two.**

Table 11 brings together the variables collected at national scale and associates them with the driver used for their downscaling:

Table 11. Downscaling driver association

Downscaling driver	Module	Variables at national level
GDP	Economy	Labour Share
		Compensation of employees
		GFCF
		Capital stock
		Net exports
		Depreciation
		Gross operational surplus
		Households & Government Consumption
Population	Energy	Total Final Energy Consumption by sector and type
	Energy	Total Final Consumption by sector and type – Household validation
	Water	Water Consumption per capita
Area	Energy	Coal Reserves

4.2.1. GDP downscaling

Reducing GDP data resolution is one of the most complex issues to face when developing a model, especially in those cases in which even sub-national data accessibility (regional, county-level...) becomes a problem. This is the situation of many African countries, whose governments or national statistical departments only release country-level economic information.

Working with more specific GDP values, e.g. at city or province level, ensures higher accuracy in the downscaling process because the more general the starting data, the larger the error could be and the easier it is to miss relevant information through it. Fortunately, the South African Department of Statistics (Stats SA, 2023) produces annual provincial reports from which it is possible to derive provincial GDP time series, divided by productive sector.

Knowing the GDP of Mpumalanga, the province where the IUWMA is located, it is now necessary to downscale it to the basin level in order to separate the contribution of the IUWMA from the rest of the province. This difficult exercise can be addressed from different perspectives, but for ONEPlanET project (Wang & Sun, 2022)'s approach has been followed.

This paper proposes to use a disaggregation method that combines gridded population and night-time light (NTL) images to obtain a proxy geolocated GDP. The method, also employed by (Eberenz et al., 2020; Zhao et al., 2017), assumes a proportionality between GDP, NTL and population per grid by creating a parameter called *LitPop*, measured in digital numbers (DN). *LitPop* calculation ensures that clusters with luminosity but no counted population and vice versa (populated but not illuminated) are not overlooked. *LitPop* can be calculated following **Equation 1**:

$$LitPop = \begin{cases} PopifLit = 0 \\ Lit \cdot PopifLit > 0 \wedge Pop > 0 \\ LitifPop = 0 \end{cases} \quad \text{Equation 1}$$

where *Pop* is the population count in each grid and *Lit* is the NTL measured in each grid.

These gridded NTL and population data can be obtained from PopClusters (Khavari, 2020), which gathers NTL, population, area, access to electricity and type of interface (rural, peri-urban or urban) for sub-Saharan Africa in 2016. The population comes from Global Human Settlements (GHSL)¹ and High-Resolution Settlement Layer (HRSL)², while NTL is obtained from the NTL map³.

Since PopClusters contains gridded data at the national level, it is necessary to extract only the part belonging to Mpumalanga and this is done by using QGIS Software.

Once *LitPop* is calculated for each grid, it can be used to obtain a GDP proxy in each area by applying **Equation 2**:

¹ <https://human-settlement.emergency.copernicus.eu/datasets.php>

² <https://www.ciesin.columbia.edu/data/hrsl/>

³ <https://earthobservatory.nasa.gov/features/NightLights>

$$GDP_{grid} = \frac{GDP_i}{SLP_i} \cdot LitPop_{grid} \quad \text{Equation 2}$$

where GDP_i represents the known GDP of a given area “ i ” (Mpumalanga in the case of IUWMA, as it is the most precise data available) in millions of constant 2015 USD, SLP_i is the sum of the LitPop data within the same area “ i ”, and $LitPop_{grid}$ refers only to the area of interest (IUWMA).

To this end, LitPop is calculated for both the IUWMA and the whole of Mpumalanga and, using the 2016 provincial GDP, the watershed GDP can be obtained as explained in **Equation 3**:

$$IUWMAGDP(2016) = \sum GDP_{grid_{IUWMA}} = \sum \frac{GDP_{Mpuma.(2016)}}{SLP_{Mpuma.}} \cdot LitPop_{grid_{IUWMA}} \quad \text{Equation 3}$$

The main inputs and results involved in this downscaling are shown below in **Table 12**:

Table 12. Variables involved in GDP downscaling

Variable	Type	Units	Value	Source
Mpumalanga GDP for 2016	Input	2015 constant USD millions	26,217.96	(Stats SA, 2023)
Mpumalanga NTL	Input	DN	-	(Khavari, 2020)
Mpumalanga Population	Input	People	-	(Khavari, 2020)
IUWMA LitPop	Intermediate calculation	DN	Depends on the grid	-
Mpumalanga SLP	Intermediate calculation	Digital numbers	108,338,833.1	-
IUWMA GDP	Result	2015 constant USD millions	9,936.28	-

4.2.2. Population and Area data processing

As aforementioned in **Sub-section 4.2.1**, gridded population and area data are accessible through PopClusters (Khavari, 2020), but as with GDP, a proper treatment is required to adequate the scale to the CS. Therefore, the clusters located in the area are extracted with QGIS and their data are aggregated to produce the basin-level information.

Alongside these two main variables, the task is carried out in parallel for other relevant factors broken down by type of interface, e.g. population at each interface, access to electricity, etc. **Table 13** presents the processed data for the Inkomati-Usuthu Case Study:

Table 13. Processed data for the IUWMA

Variable		Units	Value
Total Area		km ²	36,193
Total Population		people	2,095,361
Population	Rural	people	200,840
	Peri-urban		478,755
	Urban		1,415,766
Inhabited Area	Rural	km ²	453.72
	Peri-urban		422.17
	Urban		865.86
Population with electricity access	Rural	People	130,594
	Peri-urban		478,425
	Urban		1,415,681
Share of total population with access to electricity		%	95.9
Mean Luminance	Rural	DN	0.66
	Peri-urban		6.12
	Urban		17.65

4.2.3. GDP-based downscaling processes

GDP is the cornerstone of most of the downscaling processes carried out as economy practically performs as a point of interconnection among the different modules, especially by calculating resource intensities to assess their availability or scarcity (water, land, energy, labour, etc.).

GDP is the base for downscaling most of economic variables, which are mainly country-wide and reported in economic review bulletins. As it is almost impossible to directly associate these economic variables to such a "small" area as the IUWMA, the downscaling process applied consists of assuming a proportional correlation between GDP and the rest of the variables, i.e. if the IUWMA GDP represents a certain share of the total South African GDP, the other economic parameters in the basin will maintain the same ratio with national values.

Some variables from the economy module are dimensionless (labour share, capital compensation or TLSP) and, due to the complexity of obtaining sub-national level data, they are assumed to be homogenous across the country. If, at some point, more accurate datasets for these parameters emerge, they will be updated to improve the database quality.

Equation 4 presents the method followed to downscale national economic variables:

$$EcVar_{CS} = \frac{GDP_{CS}}{GDP_{nat}} \cdot EcVar_{nat} \quad \text{Equation 4}$$

where *nat* refers to the national level and *EcVar* is the economic variable to downscale in each case.

In Addition, the sectorally disaggregated GDP can be further utilized to downscale Total Final Energy Consumption (TFEC) and energy demand by productive sector, drawing from national

energy balance data available through databases such as the International Energy Agency (IEA)⁴ and the African Energy Commission (AFREC)⁵.

4.2.4. Area-based downscaling processes

The only variable that has been downscaled per se through area has been coal reserves, by direct association of the ratio between the surface area of the province and the surface area of the CS. However, and as explained in more detail in **Section 4.3**, this downscaling method has been mainly used as validation for other downscales, such as GDP and energy demand from agricultural sector.

4.2.5. Population-based downscaling processes

Gridded population has mainly been used in the other two CS encompassing several countries to, when downscaling national variables, give a weighted value to each country according to the proportion of inhabitants contributing to the population of the river basin. Thus, those variables strongly linked to population density have been weighted with these shares in order to better capture the importance or the greater development of one country over another.

Furthermore, as in the case of the area, population downscaling has also served as a validation method for the downscaling of GDP and household consumptions.

4.3. Data validation

In general, data validation is a very difficult task, especially when there is so much uncertainty in the case studies. Validation of the GDP downscaling results presents a complex challenge due to the absence of official or accurate subnational data for direct comparison. To address this, three alternative validation approaches have been employed: (i) GDP estimation based on population downscaling, (ii) GDP estimation based on area downscaling, and (iii) province-level GDP disaggregation and comparison with official statistics.

(i) GDP Estimation Based on Population Downscaling: Rather than applying the LitPop method (which combines population and NTL data), this approach relies solely on the population residing within the case study area. It assumes that the share of the province's population living in the CS is proportional to the share of GDP, as expressed in **Equation 5**:

$$GDP_{CS} = \frac{Pop_{CS}}{Pop_{prov}} \cdot GDP_{prov} \quad \text{Equation 5}$$

(ii) GDP Estimation Based on Area Downscaling: following a similar logic to the population-based method, this approach uses the spatial extent of the CS in relation to the entire province, under the assumption that GDP is uniformly distributed by area. This is illustrated in **Equation 6**:

⁴ <https://www.iea.org/countries/>

⁵ <https://au-afrec.org/data-statistics-energy-balances>

$$GDP_{CS} = \frac{Area_{CS}}{Area_{prov}} \cdot GDP_{prov} \quad \text{Equation 6}$$

(iii) Province-Level GDP Disaggregation and Comparison with Official Reports: in provinces where official GDP data is available, the GDP of the portion of the province outside the CS can be downscaled. The result is then compared to the sum of the estimated GDPs for the CS and the non-CS areas, as shown in **Equation 7**:

$$GDP_{prov} = GDP_{CS} + GDP_{non-CS} \quad \text{Equation 7}$$

Table 14 shows the area and population corresponding to the IUWMA, Mpumalanga and the remaining part, as well as the results obtained by the three validation methods and the official GDP of the province. As can be seen, the total error of the GDP obtained by LitPop downscaling for Mpumalanga with respect to the official one is only 2.6%, which is satisfactory:

Table 14. Area, population, official GDP and downscaled GDP of IUWMA, non-IUWMA and Mpumalanga in 2016

	Area (km ²)	Population	Official GDP (2015 M\$)	LitPop GDP	Population GDP	Area GDP
IUWMA	36,193.1	2,095,361	-	10,384.42	12,351.19	12,403.71
Non-IUWMA	40,301.9	2,352,473	-	15,151.42	13,866.76	13,814.24
Mpumalanga	76,495.0	4,447,834	26,217.96	25,535.84	-	-

The TFEC is validated by means of two different approaches. On the one hand, the shares by sector in the basin are compared with the national percentages, and on the other hand, the estimated consumption for households and agriculture are compared with those that would be obtained by downscaling them by population and area, respectively. **Table 15** compares the TFEC broken down by sector both at CS and national level:

Table 15. TFEC by sector for IUWMA and South Africa

	IUWMA		South Africa			
	TFEC	Sectoral share	IEA		AFREC	
			TFEC	Sectoral share	TFEC	Sectoral share
Industry	8.89	38.57%	263.46	38.56%	282.14	37.61%
Mining and Quarrying	4.76	20.74%	62.71	9.18%	77.76	10.36%
Agriculture, Forestry and Fishing	1.15	4.97%	23.72	3.50%	23.52	3.14%
Transport	2.84	12.31%	187.71	27.47%	179.60	23.94%
Commercial and Public Services	1.41	6.10%	52.04	7.6%	58.62	7.81%
Households	4.01	17.39%	93.63	13.72%	128.50	17.13%
Total	23.06	100%	683.27	100%	750.13	100%

Most of the sectoral shares match between the scales, with the only significant differences being in mining and transport. The reason for this divergence could be due to the fact that the IUWMA is a predominantly mining region that hosts most of the country's coal mines. This high endowment of mines translates into a higher GDP.

5. Metadata considerations

Metadata is an essential component of any database, especially for those of a public nature, because it allows for correct transparency and consistency of the data collection processes. Adequate metadata inclusion ensures an easy traceability of the data used in the Project, facilitating its consultation and validation by those who use the model in future stages, as well as a potential update of the values if their sources of origin get updated in new versions.

5.1. Structure and definitions

ONEPlanET's database includes three different classes of metadata to cover all the information that an external user might need to have a complete view and understanding of what it is been used: (i) definition of the variable; (ii) data source; and (iii) calculation methodology for those needed and measurement units. These three categories are briefly explained below:

- **Definition of the variable:** even though some variables are easily understandable, others may become a bit confusing for those without a proper background. Thus, each parameter is explained concisely, both in the database and within the model.
- **Data source:** each variable contains information on the specific dataset from which it has been collected, with the corresponding citation and link to the online repository. This is providential since a lot of data repositories are continuously being improved and updated, releasing new versions with greater accuracy and more up-to-date information.
- **Calculation methodology:** many of the data that make up the database are not directly available anywhere, either because some variables need to be calculated by using equations, or because a downscaling process has been applied to get the required granularity. In both cases, the procedure to obtain the numerical values is included, briefly detailing the formula and parameters used.

There are several methods to implement the metadata in the database, but in order to keep the Excel file simple to use, all metadata appears in the columns next to the “variable name” cell. These cells can be modified, consulted and deleted with ease by any user.

6. Challenges, barriers and solutions

Data searching always poses complexities due to the accessibility to open yet reliable datasets, regardless of the field of work or the scale of application. Nevertheless, these issues become even greater when data topic is still poorly developed, as is the case in this project.

This set of problems occurred during the iterative execution of data mapping and modeling has resulted in unexpected delays that have had an impact on the proper development of some of the regional models built in WP3 and, consequently, of the entire project.

6.1. Main challenges faced during collection

The search for specific data at sub-national level is often really complex, but especially when targeting developing countries. ONEPlanET not only focuses on African countries whose national statistics departments lack consistency, transparency and accessibility, but works at the river basin level, resulting in increased modelling complexity. These basins do not represent “official” or administrative boundaries, but physical and hydrographic systems, so there is no specific data for them as a whole.

To overcome these challenges, ONEPlanET is aiming to **enhance** data accessibility, interoperability and usability and **improve information flows** across different scales and levels. This challenge was already taken up from the beginning of the project and was expected to be overcome with the usage of **geolocated data**. However, many variables essential for the correct development of the models are not collected in any geolocated database, which makes it crucial to consult national sources for subsequent downscaling. The main barrier in this step should be the intrinsic complexity of downscaling, but in some cases, it has been harder to find national information than to process it (in particular in the Songwe and Bani CSs).

Finally, other challenges faced have been the compilation of feedback and accurate data initially expected from stakeholders in the Songwe and Bani River Basin (which is often more difficult for outsiders to access), as well as the approach to gather and downscale data when the river basin is transboundary, because it is necessary to consider the different influence and relevance of the involved nations.

6.2. Solutions found and lessons learnt in the process

Given that the modelled river basins' statistics are not available on any specific data systems, the best option is, if possible, to employ GIS data and process it to work at the basin scale. When it has been necessary to rescale the information, it has been decided to try to find several different sources to validate the data and to be able to compare the results obtained after downscaling. When data was not available from either of the two sources, it has been agreed with the modellers to slightly modify the models to fit the availability of information for each river basin.

7. References

- Burek, P., Satoh, Y., Kahil, T., Tang, T., Greve, P., Smilovic, M., Guillaumot, L., Zhao, F., & Wada, Y. (2020). Development of the Community Water Model (CWatM v1.04) – a high-resolution hydrological model for global and regional assessment of integrated water resources management. *Geoscientific Model Development*, 13(7), 3267–3298. <https://doi.org/10.5194/gmd-13-3267-2020>
- Copernicus Climate Change Service. (2019). *Land cover classification gridded maps from 1992 to present derived from satellite observations* [Dataset]. ECMWF. <https://doi.org/10.24381/CDS.006F2C9A>
- Eberenz, S., Stocker, D., Rösli, T., & Bresch, D. N. (2020). Asset exposure data for global physical risk assessment. *Earth System Science Data*, 12(2), 817–833. <https://doi.org/10.5194/essd-12-817-2020>
- Gilbert, M., Cinardi, G., Da Re, D., Wint, W. G. R., Wisser, D., & Robinson, T. P. (2022). *Global goats distribution in 2015 (5 minutes of arc)* (Version 1.0, pp. 303929, 1950394, 354205, 5861552, 223557, 2536746, 1175984, 9741599) [Image/png,image/tiff,image/png,image/tiff,image/png,text/html,image/tiff,image/tiff]. Harvard Dataverse. <https://doi.org/10.7910/DVN/YYG6ET>
- Global Energy Observatory (GOE), Google, KTH Royal Institute of Technology in Stockholm, Enipedia, & World Resources Institute (WRI). (2021). *Global Power Plant Database v1.3.0* [Dataset]. <http://resourcewatch.org/>
- Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S. A., Tyukavina, A., Thau, D., Stehman, S. V., Goetz, S. J., Loveland, T. R., Kommareddy, A., Egorov, A., Chini, L., Justice, C. O., & Townshend, J. R. G. (2013). High-Resolution Global Maps of 21st-Century Forest Cover Change. *Science*, 342(6160), 850–853. <https://doi.org/10.1126/science.1244693>
- Harris, N. L., Gibbs, D. A., Baccini, A., Birdsey, R. A., De Bruin, S., Farina, M., Fatoyinbo, L., Hansen, M. C., Herold, M., Houghton, R. A., Potapov, P. V., Suarez, D. R., Roman-Cuesta, R. M., Saatchi, S. S., Slay, C. M., Turubanova, S. A., & Tyukavina, A. (2021). Global maps of twenty-first century forest carbon fluxes. *Nature Climate Change*, 11(3), 234–240. <https://doi.org/10.1038/s41558-020-00976-6>
- International Food Policy Research Institute. (2019). *Global Spatially-Disaggregated Crop Production Statistics Data for 2000 Version 3.0.7: Spatial Production Allocation Model (SPAM) 2000* (Version 2.1). Harvard Dataverse. <https://doi.org/10.7910/DVN/A50I2T>
- International Food Policy Research Institute (IFPRI). (2019). *Global Spatially-Disaggregated Crop Production Statistics Data for 2010 Version 2.0: Spatial Production Allocation Model (SPAM) 2010* (Version 4.1). Harvard Dataverse. <https://doi.org/10.7910/DVN/PRFF8V>
- International Food Policy Research Institute (IFPRI). (2020). *Spatially-Disaggregated Crop Production Statistics Data in Africa South of the Sahara for 2017: Spatial Production Allocation Model (SPAM) 2017 in Sub-Sahara Africa (SSA)* (Version 3.1). Harvard Dataverse. <https://doi.org/10.7910/DVN/FSSKBW>

International Food Policy Research Institute (IFPRI). (2024). *Global Spatially-Disaggregated Crop Production Statistics Data for 2020 Version 1.0* [Dataset]. Harvard Dataverse. <https://doi.org/10.7910/DVN/SWPENT>

International Food Policy Research Institute (IFPRI) & International Institute For Applied Systems Analysis (IIASA). (2016). *Global Spatially-Disaggregated Crop Production Statistics Data for 2005 Version 3.2: Spatial Production Allocation Model (SPAM) 2005 (Version 9.3)*. Harvard Dataverse. <https://doi.org/10.7910/DVN/DHXBXX>

Khavari, B. (2020). *PopClusters* [Dataset]. Mendeley. <https://doi.org/10.17632/Z9ZFHZK8CR.6>

Lehner, B., Beames, P., Mulligan, M., Zarfl, C., De Felice, L., Van Soesbergen, A., Thieme, M., Garcia De Leaniz, C., Anand, M., Belletti, B., Brauman, K. A., Januchowski-Hartley, S. R., Lyon, K., Mandle, L., Mazany-Wright, N., Messenger, M. L., Pavelsky, T., Pekel, J.-F., Wang, J., ... Higgins, J. (2024). The Global Dam Watch database of river barrier and reservoir information for large-scale applications. *Scientific Data*, 11(1), 1069. <https://doi.org/10.1038/s41597-024-03752-9>

Lehner, B., & Grill, G. (2013). Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15), 2171–2186. <https://doi.org/10.1002/hyp.9740>

Messenger, M. L., Lehner, B., Grill, G., Nedeva, I., & Schmitt, O. (2016). Estimating the volume and age of water stored in global lakes using a geo-statistical approach. *Nature Communications*, 7(1), 13603. <https://doi.org/10.1038/ncomms13603>

Stats SA. (2023). *Provincial gross domestic product: Experimental estimates, 2013-2022*. Statistics South Africa.

The World Bank. (2024). *Global Wind Atlas* [Dataset]. <https://globalwindatlas.info/en/>

The World Bank. (2025). *Global Solar Atlas 2.0* [Dataset]. <https://globalsolaratlas.info/map>

Wang, Xinyu; Meng, Xiangfeng; Long, Ying (2022). Projecting 1 km-grid population distributions from 2020 to 2100 globally under shared socioeconomic pathways. figshare. Dataset. <https://doi.org/10.6084/m9.figshare.19608594.v2>

Wang, T., & Sun, F. (2022). Global gridded GDP data set consistent with the shared socioeconomic pathways. *Scientific Data*, 9(1), 221. <https://doi.org/10.1038/s41597-022-01300-x>

WorldPop. (2020). *Global 1km Population total adjusted to match the corresponding UNPD estimate* [Dataset]. University of Southampton. <https://doi.org/10.5258/SOTON/WP00671>

Zhao, N., Liu, Y., Cao, G., Samson, E. L., & Zhang, J. (2017). Forecasting China's GDP at the pixel level using nighttime lights time series and population images. *GIScience & Remote Sensing*, 54(3), 407–425. <https://doi.org/10.1080/15481603.2016.1276705>